WORK COMPLETED BY

SOHAM PHULARE

LiH | Aer Estimator | Configuration

Objective: Establishing a Resource-Efficient VQE Pipeline

- **Simulation Backend:** Aer Estimator (Noise-free benchmarking).

- **Dimensionality Reduction:** Utilized ActiveSpace Transformer (2e, 2o) and Tapered Qubit Mapper to reduce Hilbert space dimension, minimizing memory footprint.

- **Optimization Logic:** Implemented **SPSA, ADAM** and **Gradient Descent** to compare between optimizers to check for the one showing the best behaviour.

- **Execution Control:** Integrated **Simple_Delta_Checker** for **SPSA** for early-exit convergence $\Delta E < 10^{-3}$, preventing redundant compute cycles.

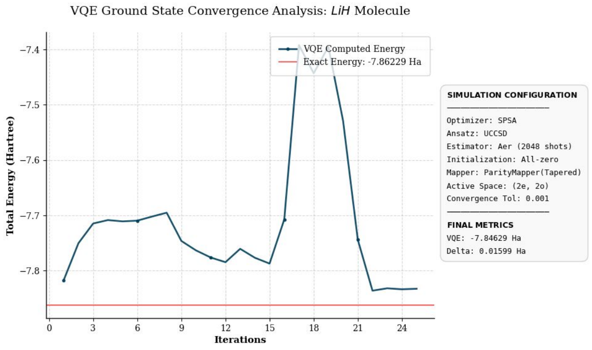
LiH | Aer Estimator | Effect of initialization

Objective: Benchmarking effect of initialization

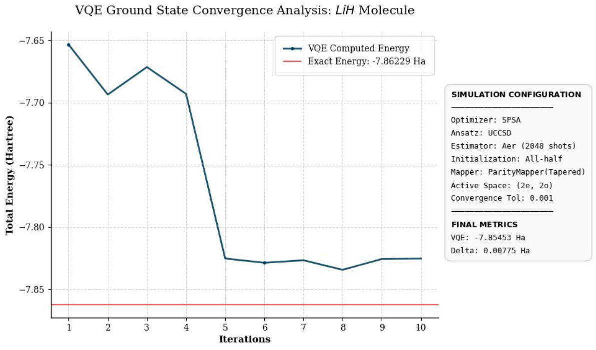
- **Comparative Performance:** Evaluated Zero vs. Half vs. One vs Random on LiH.

- **Initialization Strategy:** Validated that **Hartree-Fock (HF)** initialization reduces iteration-to-convergence count compared to random initial states, optimizing "Warm Start" performance.

- **Conclusion:** Zero initialization causes the energy to not be damped; meanwhile small initialization (half or one) causes the damping of energy to be better



Time taken: 21.104s

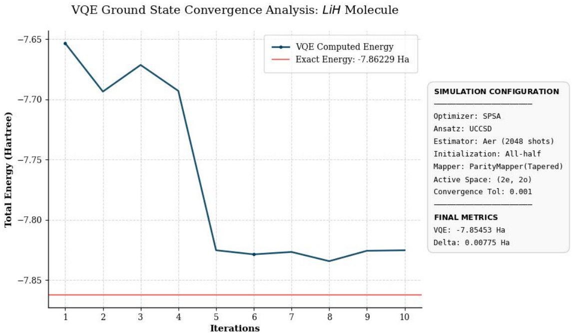


Time taken: 17.869s

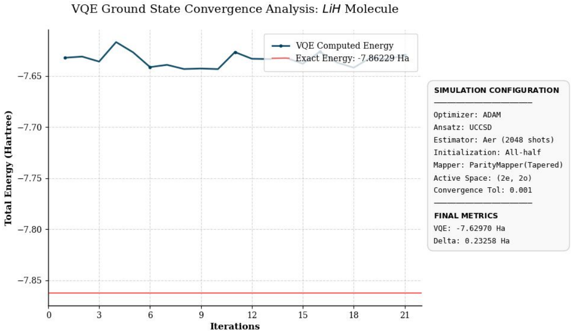
LiH | Aer Estimator | Effect of optimizer

Objective: Benchmarking effect of optimizer

- **Comparative Performance:** Evaluated SPSA vs ADAM vs Gradient Descent on LiH.
- **Optimization Strategy:** Validated that **SPSA** effectively handles stochastic noise, requiring fewer measurements per iteration than standard gradient-based approaches.
- **Conclusion:** SPSA provides significantly better convergence behavior than the ADAM optimizer, successfully minimizing the energy toward the exact ground state.



Time taken: 17.869s

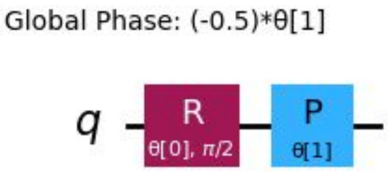


Time taken: 22.654s

H2 Quantum Run Configuration

Objective: Hardware Validation & Transpilation

- **Chemistry Model:** ROKS with 6-31g basis set.
- **Dimensionality Reduction:** Frozen core, (2e, 2o) active space, and Tapered Parity Mapper exploiting symmetry to reduce the system to a highly efficient 1-qubi circuit.
- **Hardware-Efficient Ansatz:** *EfficientSU2* (see diagram). The low gate depth minimizes hardware decoherence impacts.
- **Execution Pipeline:**
    - *qiskit_ibm_runtime* (v0.45.1) with Estimator V2 for seamless local-to-QPU scaling.
    - Hardware-aware transpilation matched to IBM QPU topology.
    - SPSA optimizer (maxiter=100) for stochastic noise resilience.



H2 | SV vs Aer vs QPU Run | Results

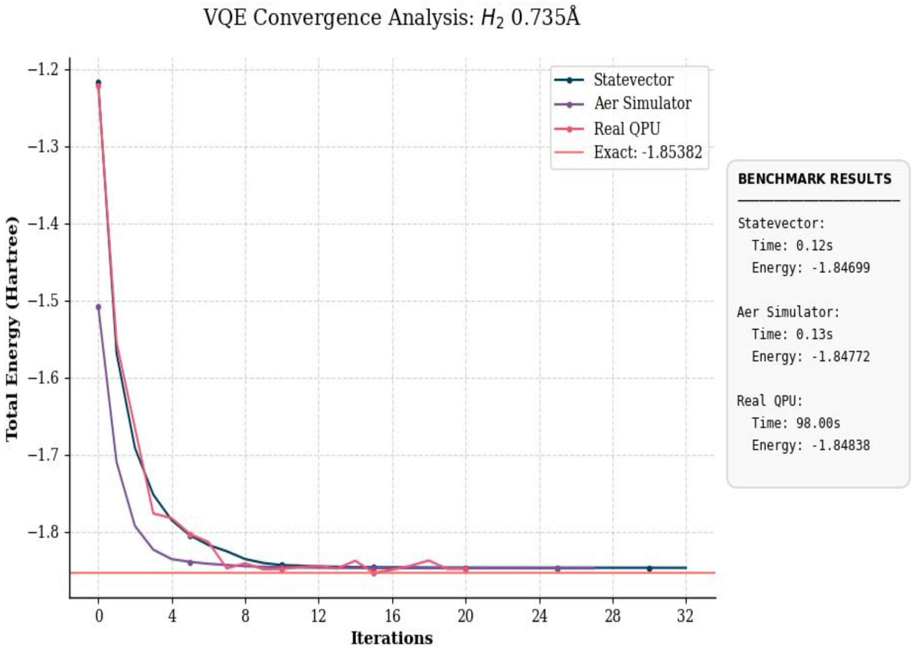
Objective: Scaling Analysis & Bottleneck Diagnostics

- **Diagnosis:** Identified a **Circuit Depth vs. Coherence Time** bottleneck. High gate-count for UCCSD Ansatz exceeded the hardware execution window.

- **Hardware Success:** Achieved ground state energy for H_2 on QPU, successfully verified the execution pipeline.

- **Time Results:** The time taken for QPU also includes the queue time on the IBM Quantum cloud, transpilation overhead, and network latency.

- **The Scaling Bottleneck:** H_2O (2e, 6o) hardware attempts resulted in "Time Limit Exceeded" errors.



Note: Exact energy calculated using FCI